

## Review

## Modeling tauopathy: a range of complementary approaches

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**Abstract**

The large group of neurodegenerative diseases which feature abnormal metabolism and accumulation of tau protein (tauopathies) characteristically produce a multiplicity of cellular and systemic abnormalities in human patients. Understanding the complex pathogenetic mechanisms by which abnormalities in tau lead to systemic neurofibrillary degenerative disease requires the construction and use of model experimental systems in which the behavior of human tau can be analyzed under controlled conditions. In this paper, we survey the ways in which in vitro, cellular and whole-animal models of human tauopathy are being used to add to our knowledge of the pathogenetic mechanisms underlying these conditions. In particular, we focus on the complementary advantages and limitations of various approaches to constructing tauopathy models presently in use with respect to those of murine transgenic tauopathy models.

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Ever since the identification of tau protein as the primary component of the neurofibrillary tangles (NFTs) of Alzheimer's disease (AD), the biology and pathobiology of human tau protein has been a major focus of research interest in the field of neurodegenerative disease. The importance of tau was emphasized by the discovery several years ago that mutations in the tau coding sequence are tightly linked to several familial neurofibrillary degenerative diseases (collectively called frontotemporal dementias linked to chromosome 17, or FTD-17). This finding established tau protein as a central factor in the pathogenesis of a large group of neurofibrillary degenerative diseases (NFDs) including AD [1–6]. Much of the recent activity in the field of tau pathobiology has been directed at the creation of transgenic (tg) mouse models in which wild-type or mutant human tau isoforms have been overexpressed under the control of a variety of promoters to produce neurofibrillary pathology. Recent murine tg models of tauopathy have been reviewed extensively elsewhere [7–9], and while the contributions of non-murine tauopathy

models have been noted in some recent reviews [8,10], such models and their potential uses have attracted relatively little attention. This review will focus on the contributions of non-murine model systems to our understanding of tau-induced and tau deposit-related NFDs in humans, centering on the cellular and systemic consequences of the intracellular accumulation and modification of human tau protein in neurons. Special emphasis will be placed on the complementary nature of information derived from cellular and non-murine tg models of tauopathy with respect to that provided by tg mouse models.

An understanding of the role played by abnormalities of tau metabolism in human tauopathies (and which must be replicated in any successful cellular or whole animal tauopathy model) must ultimately be based on a knowledge of the normal cellular functions of tau, many of which are well understood and are briefly summarized below. This discussion will be accompanied by a consideration of (a) “abnormal” characteristics acquired by tau during the course of some or all tauopathies and (b) pathological events associated with intracellular tau accumulation which may play critical roles in the cytopathogenesis of tauopathy, and which therefore are attractive targets for modeling in both in vitro and in vivo systems. A summary of the hypothetical

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basis informing most recent approaches to generating most tauopathy models (the “Tangle Hypothesis”) will then be presented (see Fig. 1). The rest of the paper will be devoted to a comparative discussion of the salient features and technical advantages and disadvantages of a variety of cell-free, cellular and whole animal models which attempt to recreate the pathological behavior of tau seen in human tauopathies. A summary of the characteristics of these systems as tauopathy models is provided in Table 1.

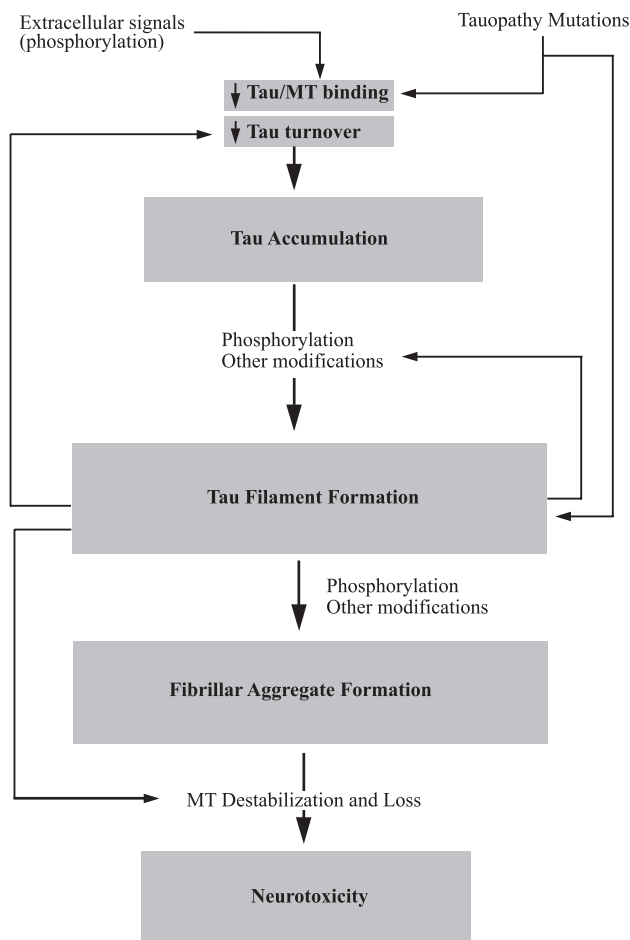


Fig. 1. A schematic diagram of the “Tangle Hypothesis” of tau-induced neurofibrillary degeneration summarizes the relationships between abnormal intracellular htau accumulation (due to a pathological failure of normal htau turnover in human tauopathies, and to htau overexpression in ABC and murine transgenic tauopathy models) and the resulting accumulation, aggregation, phosphorylation and acquisition of toxicity by tau which ultimately leads to neuronal degeneration and death. The core features of the hypothesis (shown in shaded boxes and connected with heavy arrows) provide the overall conceptual framework that serves as a point of departure for most tauopathy-related modeling approaches presently in use. Note that intracellular tau accumulation results either from abnormalities in extracellular signalling pathways (as is likely in AD) or via the abnormal MT binding properties and/or fibrillogenic changes caused by tauopathy mutations either in the coding sequence of human tau or in introns of the tau gene which affect the proportions of tau isoforms expressed in particular cells. Since (according to the Tangle Hypothesis) the key cytopathogenic events in most or all tauopathies are entirely the result of the abnormal tau accumulation, tauopathies can be modeled simply by overexpressing human tau in the target cell or organism.

## 1. Normal and pathological characteristics of human tau protein

Tau is best known as a microtubule-associated protein (MAP) which co-purifies with tubulin when the latter is put through a series of assembly/disassembly cycles [11–13]. It appears as a set of six isoforms which are differentially spliced from transcripts of a single gene located on chromosome 17 [14–16]. Each tau isoform contains a series of 3/4 tandem repeat units (depending on the isoform) which bind to MTs and serve to stabilize them [17,18]. The MT binding repeat region of tau is flanked by serine/threonine-rich regions which can be phosphorylated by a variety of kinases and which are associated with tau hyperphosphorylation in AD and other tauopathies [19–24]. The phosphorylation of these sites plays a critical role in the regulation of tau/MT interactions. Under normal circumstances, the MT binding repeats plus their flanking regions, which make up the central and C-terminal portions of all tau isoforms, are thought to be in close proximity with an MT, while the N-terminal domain of tau is thought to project away from the MT, and may interact with other cellular elements, including the plasma membrane [25].

Tau is expressed largely (but not exclusively) in neurons, where it is preferentially localized to the axonal compartment [26,27]. It appears to play a particularly important role in the stabilization of axonal MTs, and may play a key role in the initial development of axonal identity and axodendritic polarity in at least some neuronal types [28,29]. In human tauopathies, tau loses its normal axonal distribution and its normal association with MTs, and aggregates in intracellular filamentous deposits whose morphology, isoform content and intracellular location may vary depending on the tauopathy in question [30–35]. This is accompanied by a progressive destabilization and loss of cellular MTs, loss of impinging synapses, and eventually by progressive disintegration of both dendrites and the axon of the affected neuron, culminating in cell death [31,36,37]. Different neuronal types and regions of the brain are characteristically affected in different tauopathies, with pathology localized mainly to neurons, glial cells, or both, depending on the condition [33–35].

Phosphorylation of the MT binding repeat flanking domains regions of tau is abnormally prominent in the tau (PHF-tau) purified from most of the pathological tau deposits (NFTs, Pick bodies, neuropil threads, etc.) that are characteristically seen in the brains of tauopathy patients [38,39]. Consequently, the factors that influence the phosphorylation state of these sites have been a focus of research over the past 10 years. During this time, a large number of mAbs which specifically recognize many of these sites in their phosphorylated state have been generated, and the kinases and phosphatases which regulate their phosphorylation states have been identified [22–24,40–42]. Some evidence suggesting that the phosphorylation states of certain of these sites, including threonine 231 [41],

serines 242 [43] and 396/404 [44], may by themselves play key roles in determining whether and how strongly tau can bind MTs. It has also become clear that most sites of serine/threonine phosphorylation in the repeat flanking regions of tau protein are phosphorylated under some normal circumstances, including fetal development [45,46] and mitosis [47,48], and that the phosphorylation state tau is normally highly dynamic [49]. By contrast, there are indications that PHF-tau may become constitutively phosphorylated in AD [50] and very likely in other tauopathies as well [51]. Moreover, while “normal” tau has no well-defined secondary or tertiary structure, it has been shown that serine/threonine phosphorylation of tau at several of the sites mentioned above or at high stoichiometric ratios can cause tau to become “stiffer” [52], less able to bind MTs [41,43,44,53] and more resistant to proteolysis [19,54,55]. The N-terminal “projection” domain of tau contains tyrosine phosphorylation sites which are phosphorylated by signal transduction related kinases, and has been shown to interact with the plasma membrane and perimembranous structures [25]. As with the kinases and phosphatases which regulate serine/threonine phosphorylation sites of the flanking region discussed above, at least one of the tyrosine kinases which phosphorylate the tau projection domain is abnormally distributed in AD [56–58].

The question of whether the phosphorylation of “abnormal” sites on tau or the phosphorylation of tau at an abnormally high stoichiometric ratio actually facilitates tau filament formation in any significant way during the pathogenesis of any or all tauopathic syndromes is as yet unresolved, although there are some recent indications that this may be so [59,60]. Much the same can be said of a variety of other abnormal modifications that have been observed in both wild-type and mutant tau protein isoforms that make up PHF-tau, which may contribute to filamentogenesis and toxicity of tau aggregates, including truncation [61–65], abnormal folding [66] ubiquitination [67], glycation [68], nitration [69] and covalent cross-linking [70,71]. Since hyperphosphorylation of the MT flanking regions of tau and/or other abnormal characteristics of the filamentous tau found in NFTs and other pathological tau aggregates is likely to play some role in the pathogenesis of tau-related NFDs in humans, they are attractive targets for modeling (see below).

## 2. Approaches to developing tauopathy models—the Tangle Hypothesis

The great majority of cellular and whole animal tauopathy models developed to date have been based on the assumption that the intracellular formation of tau aggregates is the key cytopathogenic event in human tauopathies, and that some toxic characteristic of the aggregates themselves perturbs the physiology of the affected neuron in such a way as to produce the character-

istic cellular lesions of human tauopathy. This assumption (called the “Tangle Hypothesis” here for convenience, see Fig. 1) is supported by the following observations:

- (1) Familial tauopathies caused by coding sequence or splicing mutations in the tau gene are invariably inherited in a dominant fashion, suggesting that tauopathy mutations cause a gain of some new toxic function by the mutant tau produced rather than a loss of a normal function, and that any “gained” tauopathic features of tau biology might themselves be cytotoxic or play a pivotal role in the cytopathogenesis of tauopathy.
- (2) Tau “knockout” mice have shown only relatively minor abnormalities in a limited range of neuronal types [72], suggesting that the loss of normal tau function in neurons in situ is either not of great consequence and/or can be compensated for by other MAPs during development.
- (3) Cellular and whole animal tauopathy models which reproduce the pattern of neurofibrillary degeneration typical of tauopathies invariably also generate tau aggregates [73–77], whereas models in which no aggregates are observed very rarely show overt signs of degeneration [78–80], suggesting the likelihood of a causal link between these events.

Any general hypothesis of the mechanisms by which abnormal tau biology leads to neurofibrillary degenerative disease must account for a key feature of the effects of both tau hyperphosphorylation and “tauopathy” mutations in the tau coding sequence; i.e., that both of these interfere with normal tau/MT interactions, thus producing a potential “loss of normal function” as well as (or instead of) a “gain of toxic function” character. The Tangle Hypothesis (i.e., that the deleterious consequences of abnormal tau aggregate formation are the central factors in tauopathy cytopathogenesis) is still consistent with these findings, since it is likely that the “normal” role played by tau in binding to and stabilizing MTs may be critical in sequestering tau from the cytoplasm, and thus keeping cytoplasmic concentrations of tau low enough to prevent spontaneous filament and aggregate formation in normal neurons. It should be noted that not all findings in the field of tauopathy research appear to be consistent with the Tangle Hypothesis as framed above; for instance, a familial tauopathy resulting from a mutation in tau has been described which features significant neuronal loss in the apparent absence of tau aggregate formation [81]. It is also possible that the reduced amount of tau function (caused either by the loss of one allele in FTD-17 or by the sequestration of otherwise functional tau in aggregates) may contribute to tauopathic phenotypes. That said, the Tangle Hypothesis is generally consistent with and/or supported by the vast majority of our present knowledge of human tauopathies, and serves as a broad conceptual framework for the generation of most of the tauopathy models to be considered below.

Table 1

Summary of the comparative advantages and limitations of cellular and tg models of human tauopathies discussed in the text

| Model type                       |                                                            | General advantages                                                                    | General limitations                                              | Specific examples               | Specific advantages                                                                        | References            |
|----------------------------------|------------------------------------------------------------|---------------------------------------------------------------------------------------|------------------------------------------------------------------|---------------------------------|--------------------------------------------------------------------------------------------|-----------------------|
| Cellular models/<br>Cell culture | Nonneuronal cell lines (transient and stable transfection) | Simplicity (single cell type, controllable environment)                               | Incompleteness                                                   | SF9 cells                       | Extreme simplicity (no intermediate filaments, very simple processes)                      | [99,100]              |
|                                  |                                                            | Accessibility to wide variety of manipulations and techniques                         | Dissimilarity to human neurons in situ                           | CHO, COS 3T3 cells              | Ubiquity (well studied, widely available)                                                  | [47,103]              |
|                                  |                                                            | Lack of endogenous tau                                                                |                                                                  | CHO cells                       | Aggregates produced from some tauopathy mutants                                            | [76]                  |
|                                  | Neuronal cell lines (transient transfection)               | Neuronal morphology                                                                   | Abnormal extracellular environment and                           | PC12, SY5Y                      | Ubiquity. Aggregates produced from tauopathy mutants (SY5Y)                                | [105,116]             |
|                                  |                                                            | Ease of use, accessibility                                                            | Incomplete neuronal differentiation (vs. in situ)                | N-Tera 2                        | Human origin, longevity                                                                    | [112,114]             |
| In Situ                          | Inducible, stably transfected lines                        | Differentiable                                                                        |                                                                  |                                 | Relatively full differentiation                                                            |                       |
|                                  |                                                            | Ease of use once created                                                              | Relatively laborious to produce (vs. other cultured cell models) | H4 BE(2)-M17D                   | Human origin, inducible                                                                    | [108,118]             |
|                                  | Primary culture                                            | Fully differentiated neurons                                                          | Difficult to transfect                                           | Tau KO mice                     | Filamentous tau aggregates                                                                 | [119]                 |
|                                  | In situ cellular                                           | In situ advantages (full differentiation, possibility of long term experiments)       | Relatively laborious                                             | Lamprey ABC                     | Identified cells (unlike rodent models)                                                    | [73,74, 125,126, 138] |
|                                  |                                                            | Non transgenic background (normal extracellular environment). Drug screening possible | Few cells (no biochemistry)                                      |                                 | Large neuron size permits elaborate single cell experiments (intracellular microinjection) |                       |
| Whole animal models (transgenic) | In situ multicellular                                      | In situ and non-transgenic                                                            | Same as lamprey, also: Relatively expensive                      | In situ transfected rat neurons | Greater accessibility in situ than tg models                                               | [128]                 |
|                                  |                                                            | Can model intercellular interactions on single cell level                             | No identified cells                                              |                                 | Same as lamprey, also: Phylogenetic similarity to humans                                   |                       |
|                                  | Invertebrate                                               | Inexpensive, can be produced in quantity                                              | Phylogenetic dissimilarity to humans                             | <i>C. elegans</i>               | Large animal size, future creation of tg strains is possible                               |                       |
|                                  |                                                            | Favors biochemical and genetic analysis                                               | Inaccessibility of neurons in situ (small size)                  |                                 | Simple, highly stereotyped and well studied development                                    | [129]                 |
|                                  |                                                            | Behavioral analysis possible                                                          |                                                                  | <i>Drosophila</i>               | Well developed genetics with short generations                                             |                       |
| Whole animal models (transgenic) | Lower vertebrate                                           | Less expensive than mice                                                              | Less phylogenetic similarity to humans than mice                 | Zebrafish                       | Same as <i>C. elegans</i>                                                                  | [130,131]             |
|                                  |                                                            | Vertebrate neuronal morphology                                                        |                                                                  |                                 | Well studied development                                                                   | [134]                 |
|                                  |                                                            | Greater simplicity and stereotypy at cellular and intercellular levels than mice      |                                                                  |                                 | Well studied genetics                                                                      |                       |
|                                  |                                                            |                                                                                       |                                                                  |                                 | Optical clarity                                                                            |                       |

(continued on next page)

Table 1 (continued)

| Model type                       | General advantages                                                             | General limitations                    | Specific examples                   | Specific advantages                                                                      | References         |
|----------------------------------|--------------------------------------------------------------------------------|----------------------------------------|-------------------------------------|------------------------------------------------------------------------------------------|--------------------|
| Whole animal models (transgenic) | Behavioral analysis possible                                                   |                                        |                                     |                                                                                          |                    |
|                                  | Murine                                                                         |                                        |                                     |                                                                                          |                    |
|                                  | Phylogenetic similarity to humans                                              | Very expensive to create and maintain  | Murine tg, knockout/ in models      |                                                                                          | Reviewed in [7–10] |
|                                  | Most complete and versatile whole animal models of human tauopathies available | Complexity, lack of identified neurons |                                     |                                                                                          |                    |
|                                  | Established tg technology                                                      | Limited physical accessibility in situ |                                     |                                                                                          |                    |
|                                  | Ubiquity                                                                       |                                        |                                     |                                                                                          |                    |
|                                  | Behavioral analysis possible and relevant to human disease                     |                                        | “Composite” models using murine tgs | Flexibility-can be combined with primary cell culture and regional in situ manipulations | [121,137]          |
|                                  | Final screen in animal testing of anti tauopathy therapeutics                  |                                        |                                     |                                                                                          |                    |

The general nature and flexibility of the Tangle Hypothesis is one of the reasons for its utility as a tacit point of departure for modeling tauopathy. The predictions of this hypothesis are very broad, and do not explicitly distinguish between the pathogenetic consequences of tau aggregate formation itself and those of modifications of tau (hyperphosphorylation, truncation, glycation, etc.) which are always found in the context of tau aggregates in human tauopathy patients. The Tangle Hypothesis thus provides an adequate paradigm not only for the generation of tauopathy models which feature the development of filamentous tau aggregates (until recently, a very small minority of all tauopathy models), but also for any model in which “PHF-tau”-specific tau modifications can be generated. Models which do not themselves feature the reproduction of tau aggregates can therefore be of great utility in studying and differentiating between the cellular effects of individual PHF-tau-specific tau modifications in the absence of other such modifications and of tau filament formation itself. Most major models of tauopathy presently in use are thus either tacitly or explicitly based on the Tangle Hypothesis. This includes the large number of models that are based on altering aspects of cellular functions other than tau (kinase/phosphatase expression and activity, etc.) resulting in PHF-like modifications of endogenously expressed tau. For the sake of simplicity, the following discussion will be limited to those models that are based on the overexpression of either wild-type or mutant human tau.

Each of the models discussed below is being used to address one of the following questions: (1) Why are tau aggregates toxic? (2) What is the mechanism of this toxicity? (3) Which of the many modifications of tau specific to pathological aggregates is potentially a cause as well as a consequence of tau aggregation?

### 3. Formation of tau aggregates

The characteristic aggregates of PHF-tau seen in human tauopathies can be generated from native tau protein in any of three ways: (1) from all six isoforms (or a subset of these) of essentially normal tau which may have been modified in a variety of ways (as in AD and in sporadic forms of non-AD tauopathies), (2) from essentially normal tau isoforms which have been abnormally spliced to generate abnormal isoform ratios, and (3) from abnormal tau isoforms containing point mutations within the tau coding sequence. Either of the latter two situations is sufficient to induce any of several familial tauopathies [1–5], a finding which established the central importance of tau pathobiology to the etiology of both AD and other tauopathies. This review will exclude consideration of tauopathy models directed at the second of these mechanisms (i.e., FTD-17 mutations which induce aberrant splicing of the tau gene, resulting in abnormal isoform ratios), since studies based on the use of models of this type have not as yet been published.

### 4. General types of tauopathy models

In this review, three types of tauopathy model are considered. These are (in order of increasing complexity and scope) (1) cell-free, (2) cellular and (3) whole animal transgenic (tg) models. Each of these model types offers inherent advantages and disadvantages for advancing our understanding of abnormal tau biology, and ultimately for elucidating the mechanisms of tau induced neurofibrillary disease. The advantages of simpler models are generally those of greater accessibility to manipulation and flexibility relative to the complementary advantages of completeness and more direct relevance to human disease that are the



hallmarks of the more complex whole animal models, particularly the many murine transgenic models. These complementary advantages have permitted investigators in the field to identify features of tau function *in vivo* that are potentially relevant to the etiology of human tauopathy in cellular models (usually in cultured neuronal cell lines), and then testing their relevance to the pathogenesis of human disease in whole animal models, usually in murine transgenics. In addition to these widely used systems, several other cellular and whole animal models which offer additional technical advantages in modeling tauopathy (lamprey, nematode, zebrafish, fruit fly) are discussed. These models will be considered roughly in order of increasing complexity below, and will be preceded by a brief discussion of the simplest type of model of all: the cell-free (*in vitro*) model.

### 5. Cell-free models of tau function and dysfunction

Over the past 25 years, *in vitro* studies of the properties of human tau have resulted in a detailed understanding of many aspects of both normal and abnormal tau functions. In particular, the ability of tau to modify MT stability and MT bundle formation and the ability of tau phosphorylation to alter this ability have been heavily studied *in vitro* [42,82–84]. The models used in such studies consist of purified tubulin, tau protein and the cofactors necessary for MT polymerization. More recently, various conditions and cofactors that permit tau polymerization have been identified, as well as their relevance to the cellular environment(s) in which tau aggregates form in human tauopathies considered. It has been known since the early 1990s that given a sufficient concentration of tau protein, it is possible to induce the polymerization of human tau into filaments that resemble those seen in the tauopathies and in AD [85,86]. Since that time, a number of cofactors, including heparin and related proteoglycans [87–89], fatty acids [90,91], and RNA [92], have been identified which permit the polymerization of tau under physiological or near physiological conditions of pH and electrolyte concentrations. In addition, a number of tauopathy-associated modifications have been studied and evaluated in terms of whether they might inhibit the normal turnover of tau. Finally, cell-free models have been heavily used over the past few years to study the properties of tau isoforms containing mutations that have been shown to induce familial tauopathies by comparing their properties with corresponding wild-type tau isoforms. These studies have shown that most such “tauopathy mutations” cause either an acceleration of tau filament formation relative to wild type at a particular tau concentration [93,94] or a decrease in the ability of tau to bind to and stabilize MTs *in vitro* [95].

Cell-free models give the experimenter the power to eliminate uncontrolled variables and thus tend to generate very detailed and well-defined information. However, this approach tends to limit the field of study to aspects of tau

biology which can readily be studied *in vitro*. Thus, the details of tau/MT interactions are particularly well understood because this interaction can be readily modeled in simple *in vitro* systems. Similarly, the details of tau filament assembly, including those of the formation of intermediate oligomer species [96], and the inhibition of filament formation by exogenous agents [97] have been elucidated using cell-free models. By contrast, it is likely that the mechanics of other aspects of normal and abnormal tau function, in particular those having to do with tau/membrane interactions, are far less well understood than are tau/MT interactions and tau polymerization primarily because of the difficulty of studying them adequately in cell-free systems. The major limitation of cell-free systems in modeling the normal cell biology of tau and that of tauopathy thus appears to be their incompleteness. Thus, *in vitro* modeling is clearly an essential tool for reaching a detailed understanding of the mechanisms by which tau interacts with specific cellular elements. However, more complex models involving live cells are required in order to identify and place in perspective the importance of the various points of interaction between the aggregation and modification of tau deposits and the normal physiological processes of the neurons which contain them.

### 6. Cellular models

A general virtue of cellular models is that they offer a relatively easy, accessible and flexible approach to modeling complex events that are defined at the level of a single cell. By their very nature, cellular models of diseases with complex etiology are limited and incomplete; they can replicate only that part of the disease process which develops autonomously within a given cell. However, since the defining event of almost all human tauopathies is the generation of intracellular tau aggregates, cellular systems, and in particular, cultured cell lines, are highly appropriate for modeling the cytopathology of tauopathies. Just as the highly defined and incomplete nature of *in vitro* models makes possible a detailed elucidation of the interactions between tau and other specific cellular elements, the limited scope of cellular models can be considered advantageous in that it simplifies experimental design and interpretation. Relationships between individual cellular events that are identified by manipulations of cellular tauopathy models can then be tested for their more general applicability in whole animal models.

It should be noted that not all cellular models of tau function involve cell culture; for instance, certain differentiated cell types with established advantages (size, accessibility, simplicity) for studying tau biology such as *Xenopus* oocytes have been used to quantify the ability of wild-type and mutant human tau isoforms to interact with MTs participating in a stereotyped developmental event [98]. An *in situ* cellular model of tauopathy has been developed by the author and his colleagues using identified neurons in the

ammocoete sea lamprey; this is described in detail below. For a variety of historical and practical reasons, however, most systems used to model tauopathy at the cellular level have been based on immortalized cell lines of both neuronal and non-neuronal origin, which are differentiated and cultured *in vitro*.

## 7. Non neuronal cell lines in culture

An example of a very simple system which can be viewed as a logical extension of *in vitro* studies of tau function is the insect-derived SF9 cell line, which can be transfected with constructs expressing exogenous human tau using the baculovirus transfection system. SF9 cells not only lack endogenous tau but also lack intermediate filaments, and can be induced to generate long axon-like processes by transfection with human tau [99]. These processes appear to contain little but MT bundles and tau protein, and thus are ideal for studying the effects of tau on MT spacing and stability in the context of process formation. Recently this system has also been used to examine the effects of tauopathic mutations in the tau coding sequence on tau/MT interactions *in vivo* [100].

A variety of other non-neuronal cell lines have proven useful as tauopathy models, including fibroblast lines such as 3T3 and L cells [101], Cos cells, [102] and CHO cells [76]. Such systems have traditionally been heavily used by cell biologists and have the advantages of being very well characterized and easy to use in studies using many different cell biological and biochemical techniques. The fact that tau is largely localized to the nervous system makes certain non-neuronal cells and cell lines ideal systems for examining the properties of exogenous human tau in the absence of endogenous tau. This facilitates analyses of the normal properties of human tau as a MAP, allowing for the study of the properties of each region of the tau protein via the systematic use of deletion mutants of human tau isoforms [25,103–105]. Such models have also proven particularly useful for validating the results of *in vitro* experiments in a cellular context. For example, the effects of coding sequence tauopathy mutations on both tau/MT interactions and tau filament formation have been extensively examined in cellular models, with the results of these studies being mostly congruent with similar *in vitro* studies [89,103–110]. Finally, non-neuronal cell lines are ideal for examining the effects of overexpressing exogenous human tau isoforms bearing tauopathy-specific modifications, such as pseudophosphorylation, with the aim of evaluating the cellular effects of these modifications relative to unmodified tau [111,112].

## 8. Neuronal cell lines

As is true for non-neuronal cell lines, the most popular neuronally derived cell lines for use as tauopathy models

are well-studied lines such as PC12 and SY5Y cells. A major strength of neuronally derived cell lines as tauopathy models is that they can be induced to differentiate in culture into non-mitotic cells with a neuronal phenotype. This makes it theoretically possible to study the effects of tauopathy-associated alterations on neuron-specific cellular elements (synapses, dendrites, axons, dendritic/axonal MTs) which are key targets of cytopathological events in tauopathies. These models also permit the study of interactions between overexpressed human tau and cellular functions which are particularly important in neurons, such as dynein/kinesin-mediated transport mechanisms [113]. The practical advantages that cell culture models have over other model types (ease of use, completeness on a cellular level combined with relative simplicity) have stimulated the development of several important refinements in such models. These include the use of human cell lines and elaborate culturing and differentiation conditions to produce fully differentiated, long-lived human neurons in culture [112,114–116], and the development of stably transfected human-derived cell lines in which tau expression can be induced at will by addition of a small-molecule inducer to the culture medium [117,118]. These advances now make it possible to partially overcome two serious limitations of stably transfected neuronal cell lines as tauopathy models; i.e., the time limitation imposed on experiments in which events that take years to occur are being modeled, and the disruptive effects of continuous exogenous tau expression during differentiation.

## 9. Primary culture models

Finally, primary neuron culture is an additional approach that can be taken to ensure that a fully differentiated neuronal phenotype is present, which can be a critical factor in studies of normal functions of tau, such as the developmental sorting of tau and other MAPs between axons and dendrites [28,29,119]. Primary neurons have generally not been used as tauopathy models due to the laboriousness and variability inherent in isolating them, and in particular, difficulties in transfecting them. However, primary cultures derived from transgenic mice may become a very important approach to tauopathy modeling, as discussed further below [120,121].

In summary, cell culture models of tauopathy offer a significantly more complete reconstruction of how tau interacts with cellular structures and other processes than any *in vitro* models, permitting experiments that realistically model interactions (tau-membrane, signal transduction) which cannot be reconstituted outside of intact cells in a genuinely intracellular environment. Moreover, most cell culture tauopathy models are relatively easy to create, maintain and reproduce for future experimentation.

## 10. Limitations of cultured cells as tauopathy models

While cell culture-based models of tauopathy are significantly more complete than cell-free models in reproducing the cellular conditions under which tauopathic diseases develop in human patients, the impossibility of replicating in culture the normal extracellular environment of neurons *in situ* imposes inherent limits to how closely cultured neurons can resemble neurons *in situ*. As a consequence, such systems remain incomplete models of tauopathy at the cellular level. Perhaps the most critical failing of cultured cell lines as cellular models of tauopathy is the extreme difficulty that has been encountered in the attempts to produce NFT-like tau aggregates in cultured cells. Over the past 15 years, repeated unsuccessful attempts have been made to generate tau aggregates in culture using such strategies as generating high levels of wild-type human tau expression with powerful expression vectors, treatment with relevant kinases and with broad-spectrum phosphatase inhibitors such as okadaic acid [78]. While several successful attempts at achieving this goal have very recently been published using mutant tau isoforms [76,116,118,122], or by coinubation with beta amyloid [122], the difficulty of achieving this central goal of tauopathy modeling (according to the Tangle Hypothesis) remains a critical weakness of the cell culture approach to generating cellular tauopathy models. Over the past few years, the author and co-workers have developed an *in situ* cellular model of tauopathy which overcomes the above difficulty, and which is described below.

## 11. An *in situ* cellular model of tauopathy

There are a number of requirements which must be met in order to generate a single cell tauopathy model which can be studied *in situ*, and to overcome the primary difficulty encountered with the cellular models described above. Since tau aggregations are found preferentially or exclusively in neuronal or glial cells in human tauopathies, the cell type to be studied must be derived from these cell types. Ideally the model should be neuronal, if the goal is to generate a model that can be used to characterize the “core” cytopathological events common to all tauopathies, since some tauopathic conditions such as AD do not feature glial tangles. A second requirement is that the neuron type chosen should be readily accessible to transfection with vectors that overexpress exogenous human tau. Exogenous tau expression can be accomplished in neurons *in situ* by a number of different means. These include: (a) creating a transgenic, (b) transfecting a scattered population of neurons in a juvenile animal using a viral vector [123], (c) injecting an embryonic precursor to a set of cells fated to develop as neurons [124], or (d) microinjecting vectors directly into post-mitotic, differentiated neurons. Since the features which make a cell type desirable as a tauopathy model *in situ* (i.e., non-tg

genetic background, large, accessible cell type) are also ideal for *in situ* microinjection, we chose this last approach.

We chose to model human tauopathy in a set of giant identified neurons (Anterior Bulbar Cells or ABCs) in the hindbrain of the sea lamprey, *Petromyzon marinus*. In this model, human tau is chronically overexpressed in ABCs via the intracellular microinjection of plasmids encoding a human tau isoform tagged at the N terminal with Green Fluorescent Protein (EGFP). Injected ABCs will then express tau over a period of several months, during which time the tau becomes hyperphosphorylated, forms filamentous aggregates, and induces a series of degenerative changes, including dendritic MT and synapse loss. The sequence of events induced by tau overexpression in ABCs bears a reasonable resemblance to the cytodegenerative sequence seen in AD [36,37], and neuropathological lesions seen in other tauopathies [33–35]. This model thus offers significant advantages over cell culture models in that it features the development of filamentous tau aggregates resulting in a stereotyped, stageable pattern of neurofibrillary degeneration in a fully differentiated neuron *in situ*, against a non-transgenic background. Tau-expressing ABCs are classified as belonging to Stages 1–4 of degeneration based on the extent of dendritic degeneration [125,126].

An additional advantage of the ABCs system as a tauopathy model system is its ready accessibility *in situ*. This, combined with the large size of ABCs, makes possible experiments in which physical manipulations such as microinjection can be performed repeatedly on the same cell. For example, we had previously found that chronic tau overexpression induces a significant distortion and pruning of ABC dendrites and the loss of dendritic MTs and impinging synapses [73,74]. We then determined whether this dendritic distortion was accompanied by the physical obstruction and isolation of distal dendrites by microinjecting a low molecular weight fluorescent dye (Alexa 488) into 10 EGFP-positive ABCs in intact lampreys. The movement of fluorescent label into the distal dendrites of such cells was attenuated or abolished in dendrites exhibiting distortion and/or beading (Fig. 2). This obstruction of distal dendrites by tau deposits (and/or a localized cytoskeletal collapse caused by chronic tau expression) might be expected to lead to the metabolic isolation of distal dendrites, which in turn would account for the observed distal to proximal pattern of degenerative dendritic changes. The accessibility of ABCs thus makes it possible to correlate the results of multiple manipulations in a single identified neuron, facilitating the detailed dissection of relationships between the accumulation of tau and its pathological consequences.

While the ABC model offers a number of advantages over cultured cell lines as a cellular tauopathy model, it imposes significant limitations as well. Only a few cells express tau in the brain of any one lamprey (there are six ABCs per animal), making conventional biochemical analyses impossible. In addition, the long life cycle (10



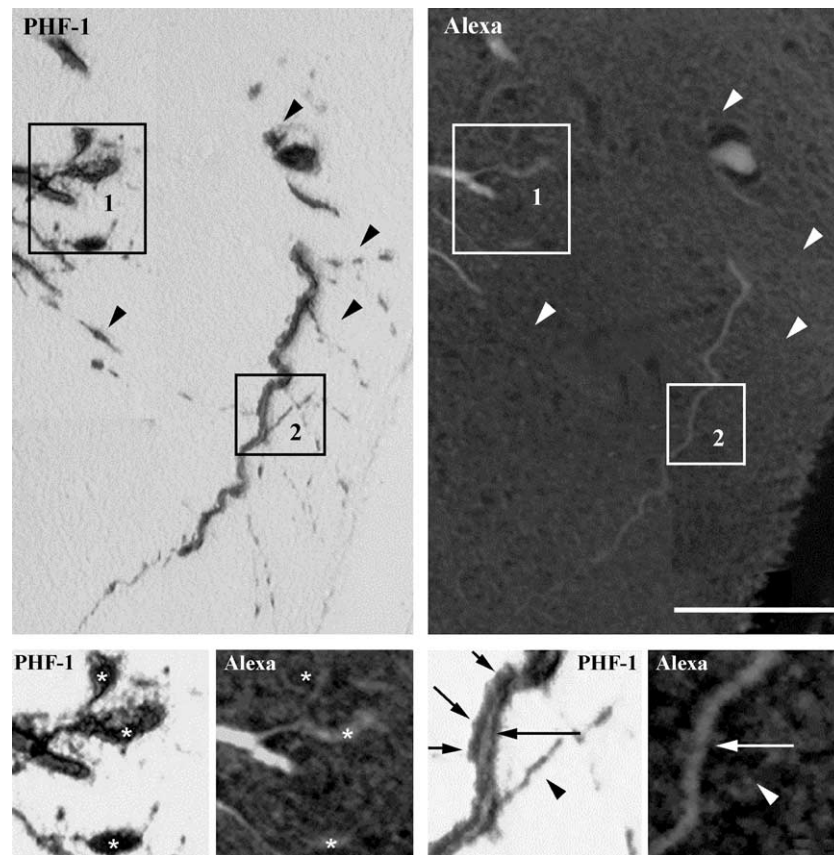


Fig. 2. Dendritic distortion and degeneration caused by chronic human tau overexpression results in the functional isolation of distal dendrites. A 10- $\mu$ m transverse section through the distal dendritic field of an ABC 66 days was microinjected with a plasmid encoding the shortest human tau isoform (tagged at the N terminal with EGFP), with expression driven by the cytomegalovirus promoter [74]. In order to test the intracellular continuity of tau expressing ABCs, the low MW dye Alexa 488 (Molecular Probes) was injected 1 h prior to fixation for 2 h in FAA [138], which in this system abolishes all EGFP fluorescence. All left-hand panels show the same section after immunostaining with the phosphotau-specific mAb PHF-1 (generously provided by Dr. Peter Davies) of dendrites from ABCs expressing EGFP-tagged tau. Right panels show the same sections prior to immunocytochemistry, showing Alexa 488 fluorescence. Magnified views of the regions marked 1 and 2 are shown at the bottom left and bottom right, respectively. We found that while Alexa 488 diffused rapidly throughout the entire dendritic trees of control cells (not shown), and relatively normal dendrites of tau expressing cells (Alexa image, inset 2), the spread of Alexa488 fluorescence into many distal dendrites in ABCs in Stage 2 or later was obstructed (asterisks, inset 1, carets, top panels and inset 2). Alexa filling of distorted dendrites shows examples of partial and complete obstruction (asterisks, inset 1) and grossly thickened regions surrounding normally appearing dendrites that label with PHF-1 but exclude Alexa (arrows, inset 2) suggest the existence of thick, perimembranous deposits of tau-positive material. Scale bar: 50  $\mu$ m.

years or more) and stability of the ammocoete lamprey brain make both genetic and developmental studies in this system impractical. The ABC model also shares with many non-murine and tg and cellular models the weaknesses of containing an endogenous tau homologue which continues to be expressed in ABCs which are also expressing human tau, and of being relatively phylogenetically distant from humans. According to the Tangle Hypothesis, however, these weaknesses should be relatively unimportant, since the normal functions of tau are not critical to tauopathy cytopathogenesis. Moreover, tau is a phylogenetically diverse protein, varying significantly among mammalian species, while the proteins with which tau has been shown to interact in tauopathies (signal transduction related kinases, PP2a, tubulin, actin etc.) are “housekeeping” proteins which vary little among animal species. On the whole, the strengths and weaknesses of lamprey ABCs as an in situ cellular tauopathy model versus those of cell culture

models epitomize the complementary nature of these models, which permit different types of manipulations at the single cell level, and result in responses that must be analyzed using quite different techniques.

## 12. Cellular versus whole animal models of tauopathy

The scope and investigative goals of whole animal tauopathy models, such as murine transgenics, are very different from (and complementary to) those of cellular models, and thus impose very different practical requirements and simplify assumptions on the experimenter, as discussed below.

In constructing and using a cellular model, the investigator tacitly assumes that the key events driving the cytopathogenesis of tauopathy are essentially cell-autonomous. The contributions of variations in cell type and the

immense variations in transsynaptic connections between different groups of neurons in situ are ignored, despite clear evidence for the importance of these factors in AD [127] and their likely importance in other tauopathies as well. The justification for doing this is contained in another assumption that is stated explicitly in the Tangle Hypothesis: i.e., that there is a single fundamental cytopathological mechanism by which all abnormalities in the coding sequence, expression and posttranslational modification of tau protein induce neuronal degeneration and death. This assumption in turn requires that the typical variations in the morphology and distribution of tau deposits seen in different tauopathies (NFTs, Pick bodies, coiled bodies, neuropil threads, etc.) are due to variations in levels of tau expression, cell type and synaptic connectivity. These simplifying assumptions are most useful if the cellular model in question is highly stereotyped (either an identified neuron or a stable cell line that has been differentiated under uniform conditions), and if manipulations of the extracellular environment are kept simple, well controlled, and highly relevant to specific aspects of tau-induced cytopathology (for example, the effects of extracellular beta amyloid oligomers on the activity of specific signal transduction-related kinases).

By contrast, the greater scope of whole animal tauopathy models requires the experimenter to consider very different issues and make different (and far more complex) simplifying assumptions in creating and using such models than would be true of a cellular model. Most whole animal models of tauopathy attempt to recreate the entire developmental sequence of neuropathological events in human tauopathy, including behavioral manifestations of disease, and such models must in addition have neuronal or glial cell populations which exhibit appropriate cytopathology. Issues that are explicitly ignored in cell models (such as the differences in relative susceptibility to tau-induced NFD that exist between various neuron types, and whether transsynaptic influences play a part in contributing to these differences) become central foci of interest in whole animal tauopathy models. A particular challenge for creating complex, ambitious whole-animal models is controlling the spatiotemporal distribution of tau expression levels among a variety of neuronal and glial populations, and thus replicating in the model the expression patterns seen in normal humans and/or in any of the various human tauopathies.

### 13. A non-transgenic, multi-cell model of tauopathy

The above discussion contrasting the scope of cellular and whole animal tauopathy models is primarily in reference to cell culture models on the one hand, versus murine tg models on the other hand, and emphasizes the large, often nonoverlapping differences between the strengths and weaknesses of each type of model and the kinds of questions that can be asked of them. The gap between the

relative scopes of these models creates potential difficulties with the interpretation of information derived from cellular models in the context of tg mice, and vice versa. One way of partially overcoming this gap has very recently been demonstrated by the production of an in situ tauopathy model in the rat [128], in which rat forebrain neurons were transfected in situ with a human tau isoform containing the P301L mutation, and thus induced to generate NFTs, neuropil threads and other cellular features of neurofibrillary degeneration. This approach shares some of the advantages enjoyed by the lamprey ABC model over murine transgenics as a cellular model discussed above (tau-induced neurofibrillary lesions are produced in individual neurons in situ in a non-tg background, large cell size and physical accessibility to further experimentation in situ). Most importantly, it also creates a multicellular model in which small-scale transcellular interactions between tau positive neurons and between transfected and non-transfected neurons can potentially be examined. Consequently, there is considerable potential for ad hoc non-tg in situ models such as this to model transsynaptic influences between neurons in human tauopathies of the cellular level in a manner presently not possible in other systems.

### 14. Transgenic non-murine whole animal models

The difficulty of integrating information derived from cellular and whole-animal models may also be overcome to some extent by using some recently developed simpler tg animal models of tauopathy which permit at least some aspects of tauopathy pathogenesis to be analyzed at the cellular level in the context of their effects on a whole animal in situ.

### 15. Invertebrate tg models

The use of non-mammalian transgenics as tauopathy models entail a tradeoff between the complexity (and by extension, completeness) of the system being used versus the expense, ease of use, and resolution of the information derived. This tradeoff is most marked in invertebrate models. Two of the best studied organisms in invertebrate neurobiology (the nematode *C. elegans* and the fruit fly *D. melanogaster*) have recently been used to create transgenic tauopathy models using normal and mutant human tau isoforms [129–131]. Both of these systems have two types of technical and scientific advantage not found in tg mice which permit experimental manipulations at the single cell level. These are (a) identified neurons with specifically defined developmental origins and stereotyped morphology, physiology and synaptic connectivity, and (b) advantages favoring extensive, rapid and detailed genetic manipulations (short generation times, small genome, simpler CNS, ability to screen for mutants). The ability to induce and analyze

genetic mutations which produce reproducible changes in individual cells of both the fruit fly and the nematode more than makes up for the small size of these organisms, which prevents physical manipulations *in situ*. The fly model may already have produced information that is of general relevance to the cytopathology of tau-induced neurodegeneration, since the overexpression of human tau isoforms in *Drosophila* results in neuronal death, but not in tau aggregates [130]. It is unclear whether this is due to a fundamental difference between neurons in the fly and in humans, such as the lack of NFs (and correspondingly large numbers of MTs) in insect neurons, or if this result is an indication of the existence of a hitherto overlooked tauopathy mechanism which does not involve tau aggregate formation. As mentioned above, familial tauopathies in which tau aggregates are absent have been described [81] and information directly relevant to this type of tauopathy might therefore be forthcoming from a fly tauopathy model. It is worth noting in this context that perturbed neurofilament distributions have been observed in neurons undergoing tau-induced neurofibrillary degeneration in tg mice, lamprey ABCs, and in human AD patients [73,75,132]. By contrast, tau overexpression failed to generate the expected number of tau aggregates in tg mouse lines lacking neurofilaments [133].

## 16. Lower vertebrate tauopathy models

Whole-animal tauopathy models derived from lower vertebrate systems might be expected to offer some, but not all, of the same benefits as the invertebrate systems described above, in exchange for greater complexity and phylogenetic similarity to humans. The single lower vertebrate system presently being used to make tg tauopathy models, the zebrafish, by and large fits this generalization [134]. In particular, this system retains some of the advantages of invertebrate tg systems, including feasibility of genetic manipulation, a well-studied, stereotyped pattern of development with embryos exhibiting a high degree of optical clarity, and the relative ease with which tg animals can be created and propagated compared to murine tg models. Most zebrafish neurons exhibit a typical vertebrate multipolar morphology, making this a particularly good system for studying the “normal” roles played by tau (and tau phosphorylation) during the development of neuronal polarity and during initial axon outgrowth. The finding that mutant human tau isoforms, as with the nematode and fruit fly, produce cytoskeletal abnormalities resembling some of those seen in human tauopathies [134] suggests that the zebrafish model may be particularly important in elucidating any developmental effects of “tauopathy” tau mutations in humans. Thus, it is likely that the greatest direct value of all of these relatively simple, non-mammalian tg systems as models for human tauopathy will be the correlation our understanding of tau

biology and pathobiology at the cellular level with our knowledge of the developmental and intercellular influences that shape the pathogenesis of tauopathies at the organismal level. Another general advantage of simpler tg models over tg mice is that they greatly decrease the amount of time, labor and expense associated with generating, maintaining and using experimental animals, which may be an important advantage in using such systems for preliminary screens of potential therapeutic agents.

## 17. Murine transgenics

The focus of the present review has been to identify ways in which information derived from non-murine tauopathy models can complement, reinforce and extend that derived from tg mouse model systems. This is because tauopathy models derived from transgenically manipulated mice are by far the most prevalent whole animal tauopathy models presently in use, and thus constitute the “gold standard” to which other tauopathy models are to be compared. There are many reasons for this pre-eminence of tg mice as tauopathy models. The laboratory mouse is the premier experimental system in biomedical research, and as such, its biology is the best understood of all mammalian (and vertebrate) animals, as is the technology for developing and manipulating murine transgenics. Murine tg models of tauopathy are also the most complete and most phylogenetically similar models to humans that can be used practically by large numbers of scientists to study human tauopathy. Over the past few years, the expression wild-type and mutant isoforms of human tau in mice has resulted in the development of NFT-like filamentous deposits in a variety of neuronal and glial populations over the life span of the mouse [75,135–137]. This has been accompanied by a broad range of behaviorally observable signs of progressive neurodegeneration (reviewed in Refs. [7–10]). A current goal in this very active field is to develop murine models which address the remaining major weaknesses in tg mouse models of tauopathy, i.e., the failure of current models to cause the degree of neuronal degeneration and death seen in human disease and their failure to reproduce human patterns of tau expression. Ultimately, tg mouse tauopathy models need to account for the characteristic distribution, morphology and isoform content of neurofibrillary lesions seen in each of the various human tauopathies, and correlate this with the nature of specific tauopathy mutations in tau. Ideally, this will involve the expression of human tau isoforms in mice null for murine tau under the control of the human *tau* promoter. This might best be attempted by using artificial chromosome techniques which permit the transfer of functional human genomic constructs to mice, resulting in normal human tau expression and splicing patterns.



The many advantages of murine tg models mentioned above have recently been expanded by the advent of “composite” modeling approaches in which tg tauopathy mice have been used in the context of other modeling approaches. Examples of such studies include the use of primary culture of neurons derived from tg mice [131], and the injection of human beta amyloid into the brains of such mice, resulting in the development of neurofibrillary pathology in the vicinity of the injection site [137]. The continual development of new approaches to producing and using tg murine models greatly increases the versatility of murine tg mice as tauopathy models, and seems likely to maintain the position of this type of model at the center of NFD research.

## 18. Conclusion

The experimental attributes required for cellular and whole animal models of NFD are inherently complementary to and quite different from one another. The systematic elucidation of the pathogenesis of tau-induced NFD requires a multilevel approach involving a detailed analysis of both the general cellular mechanisms underlying NFD at the cellular level and cell type-specific considerations and intercellular/systemic interactions responsible for the characteristic neuropathological and behavioral changes induced by individual tauopathies. Understanding the former requires that we determine how the common features of NFD are produced in a single cell type, requiring a cellular model, while understanding the latter will require whole animal models in which tau accumulation and pathology occur in the cell types that are characteristic of the human disease being considered. In particular, the use of cells derived from murine transgenics as cellular models and other “composite” approaches promise to begin addressing the greatest weakness of tauopathy models developed so far: i.e., the difficulty of using information derived from cellular and whole animal approaches in a common, context-free manner. These approaches emphasize the central importance of developing improved murine transgenic models of human tauopathies to our full understanding and treatment of these conditions. However, it seems likely that simple, non-murine tg model systems such as the worm, fly and zebrafish, and in situ cellular models such as the lamprey and rodent models discussed in this paper will remain important as tauopathy models owing to the unique perspectives that they bring to our understanding of tauopathy at both the cellular and organismal levels. The technical advantages (identified cells, accessibility, convenience, etc.) that such systems enjoy over murine transgenics and cell culture approaches, while narrow, are probably too important and fundamental to be superseded by advances in transgenic technology. Rather, investigators are increasingly likely to address particular aspects of tauopathy pathogenesis in a variety of models, using the complementary

strengths of each to assemble a complete and multilevel overview of the problem at hand. A similar approach is likely to prove useful in the identification of anti-tauopathy therapeutic agents, in which prospective drugs are screened for efficacy and then toxicity first in *in vitro* systems, then in cellular ones and finally in whole animal tauopathy models. One such compound that was recently identified as having anti-fibrillogenic activity on tau in an *in vitro* assay [97] has been shown to block tau-induced NFD in the lamprey ABC tauopathy model [126]. Thus, information derived from many different model types is likely to continue contributing to our understanding of, and ultimately our ability to treat, these devastating disorders.

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